Problem 2

Saniya_Bagwan.java

```
import java.util.*;

abstract class Employee {
   private int empId;
   private String name;
   private String email;
   private String role;

   public Employee(int empId, String name, String email, String role) {
      this.empId = empId;
      this.name = name;
      this.email = email;
      this.role = role;
   }

   public int getEmpId() { return empId; }
   public String getName() { return name; }
   public String getEmail() { return email; }
   public String getRole() { return role; }

   public void setRole(String role) { this.role = role; }

   // Abstract method
   public abstract double calculateSalary();

   public void display() {
      System.out.println("ID: " + empId + ", Name: " + name +
            ", Email: " + email + ", Role: " + role +
            ", Salary: " + calculateSalary());
   }
}

class PermanentEmployee extends Employee {
   private double baseSalary;
   private double bonus;

   public PermanentEmployee(int id, String name, String email, String role,
```

```
                    double baseSalary, double bonus) {
        super(id, name, email, role);
        this.baseSalary = baseSalary;
        this.bonus = bonus;
    }

    @Override
    public double calculateSalary() {
        return baseSalary + bonus;
    }
}

class ContractEmployee extends Employee {
    private double hourlyPay;
    private int hoursWorked;

    public ContractEmployee(int id, String name, String email, String role,
                    double hourlyPay, int hoursWorked) {
        super(id, name, email, role);
        this.hourlyPay = hourlyPay;
        this.hoursWorked = hoursWorked;
    }

    @Override
    public double calculateSalary() {
        return hourlyPay * hoursWorked;
    }
}

class InternEmployee extends Employee {
    private double stipend;
    private double performanceBonus;

    public InternEmployee(int id, String name, String email, String role,
                    double stipend, double performanceBonus) {
        super(id, name, email, role);
        this.stipend = stipend;
        this.performanceBonus = performanceBonus;
    }

    @Override
```

```
    public double calculateSalary() {
        return stipend + performanceBonus;
    }
}

// Main Class
public class Saniya_Bagwan {

    static Map<Integer, Employee> empMap = new HashMap<>();
    static Scanner sc = new Scanner(System.in);

    public static void main(String[] args) {

        int choice;

        do {
            System.out.println("\n===== SMART HR SYSTEM =====");
            System.out.println("1. Add Employee");
            System.out.println("2. Display All Employees");
            System.out.println("3. Search Employee");
            System.out.println("4. Update Employee Role");
            System.out.println("5. Delete Employee");
            System.out.println("6. Display Unique Roles");
            System.out.println("7. Exit");
            System.out.print("Enter choice: ");
            choice = sc.nextInt();

            switch (choice) {
                case 1 -> addEmployee();
                case 2 -> displayEmployees();
                case 3 -> searchEmployee();
                case 4 -> updateEmployee();
                case 5 -> deleteEmployee();
                case 6 -> displayRoles();
                case 7 -> System.out.println("Exiting...");
                default -> System.out.println("Invalid choice!");
            }

        } while (choice != 7);
    }
```

```
static void addEmployee() {
    System.out.println("\nSelect Type: 1.Permanent 2.Contract 3.Intern");
    int type = sc.nextInt();

    System.out.print("Enter ID: ");
    int id = sc.nextInt();
    sc.nextLine();

    System.out.print("Enter Name: ");
    String name = sc.nextLine();

    System.out.print("Enter Email: ");
    String email = sc.nextLine();

    System.out.print("Enter Role: ");
    String role = sc.nextLine();

    Employee emp = null;

    switch (type) {
        case 1 -> {
            System.out.print("Base Salary: ");
            double salary = sc.nextDouble();
            System.out.print("Bonus: ");
            double bonus = sc.nextDouble();
            emp = new PermanentEmployee(id, name, email, role, salary, bonus);
        }
        case 2 -> {
            System.out.print("Hourly Pay: ");
            double pay = sc.nextDouble();
            System.out.print("Hours Worked: ");
            int hours = sc.nextInt();
            emp = new ContractEmployee(id, name, email, role, pay, hours);
        }
        case 3 -> {
            System.out.print("Stipend: ");
            double stipend = sc.nextDouble();
            System.out.print("Performance Bonus: ");
            double bonus = sc.nextDouble();
            emp = new InternEmployee(id, name, email, role, stipend, bonus);
        }
```

```
        default -> System.out.println("Invalid type!");
    }

    if (emp != null) {
        empMap.put(id, emp);
        System.out.println("Employee added successfully!");
    }
}

// Display Employees
static void displayEmployees() {
    List<Employee> list = new ArrayList<>(empMap.values());

    if (list.isEmpty()) {
        System.out.println("No employees found.");
        return;
    }

    for (Employee e : list) {
        e.display();
    }
}

// Search Employee
static void searchEmployee() {
    System.out.println("Search by: 1.ID 2.Name");
    int ch = sc.nextInt();
    sc.nextLine();

    if (ch == 1) {
        System.out.print("Enter ID: ");
        int id = sc.nextInt();

        Employee e = empMap.get(id);
        if (e != null) e.display();
        else System.out.println("Not found!");

    } else if (ch == 2) {
        System.out.print("Enter Name: ");
        String name = sc.nextLine();
```

```
        boolean found = false;
        for (Employee e : empMap.values()) {
            if (e.getName().equalsIgnoreCase(name)) {
                e.display();
                found = true;
            }
        }
        if (!found) System.out.println("Not found!");
    }
}

// Update Employee
static void updateEmployee() {
    System.out.print("Enter ID: ");
    int id = sc.nextInt();
    sc.nextLine();

    Employee e = empMap.get(id);
    if (e != null) {
        System.out.print("Enter new role: ");
        String role = sc.nextLine();
        e.setRole(role);
        System.out.println("Updated successfully!");
    } else {
        System.out.println("Employee not found!");
    }
}

// Delete Employee
static void deleteEmployee() {
    System.out.print("Enter ID: ");
    int id = sc.nextInt();

    if (empMap.remove(id) != null) {
        System.out.println("Deleted successfully!");
    } else {
        System.out.println("Employee not found!");
    }
}

// Display Unique Roles
```

```
    static void displayRoles() {
        Set<String> roles = new HashSet<>();

        for (Employee e : empMap.values()) {
            roles.add(e.getRole());
        }

        System.out.println("Unique Roles: " + roles);
    }
}
```

Output:

Inserting

```
===== SMART HR SYSTEM =====
1. Add Employee
2. Display All Employees
3. Search Employee
4. Update Employee Role
5. Delete Employee
6. Display Unique Roles
7. Exit
Enter choice: 1

Select Type: 1.Permanent 2.Contract 3.Intern
1
Enter ID: 1
Enter Name: Saniya Bagwan
Enter Email: san@gmail.com
Enter Role: SDE
Base Salary: 120000
Bonus: 10000
Employee added successfully!
```

```
===== SMART HR SYSTEM =====
1. Add Employee
2. Display All Employees
3. Search Employee
4. Update Employee Role
5. Delete Employee
6. Display Unique Roles
7. Exit
Enter choice: 1

Select Type: 1.Permanent 2.Contract 3.Intern
2
Enter ID: 2
Enter Name: Payal Lokhnde
Enter Email: payal@gmail.com
Enter Role: Contracter
Hourly Pay: 1000
Hours Worked: 30
Employee added successfully!
```

```
===== SMART HR SYSTEM =====
1. Add Employee
2. Display All Employees
3. Search Employee
4. Update Employee Role
5. Delete Employee
6. Display Unique Roles
7. Exit
Enter choice: 1

Select Type: 1.Permanent 2.Contract 3.Intern
3
Enter ID: 3
Enter Name: Manasvi Gore
Enter Email: man@gmail.com
Enter Role: UI/UX designer
Stipend: 30000
Performance Bonus: 1200
Employee added successfully!
```

Displaying

```
===== SMART HR SYSTEM =====
1. Add Employee
2. Display All Employees
3. Search Employee
4. Update Employee Role
5. Delete Employee
6. Display Unique Roles
7. Exit
Enter choice: 2
ID: 1, Name: Saniya Bagwan, Email: san@gmail.com, Role: SDE, Salary: 130000.0
ID: 2, Name: Payal Lokhnde, Email: payal@gmail.com, Role: Contracter, Salary: 30000.0
ID: 3, Name: Manasvi Gore, Email: man@gmail.com, Role: UI/UX designer, Salary: 31200.0
```

Searching by ID

```
===== SMART HR SYSTEM =====
1. Add Employee
2. Display All Employees
3. Search Employee
4. Update Employee Role
5. Delete Employee
6. Display Unique Roles
7. Exit
Enter choice: 3
Search by: 1.ID 2.Name
1
Enter ID: 2
ID: 2, Name: Payal Lokhnde, Email: payal@gmail.com, Role: Contracter, Salary: 30000.0
```

Searching by Name

```
===== SMART HR SYSTEM =====
1. Add Employee
2. Display All Employees
3. Search Employee
4. Update Employee Role
5. Delete Employee
6. Display Unique Roles
7. Exit
Enter choice: 3
Search by: 1.ID 2.Name
2
Enter Name: Saniya Bagwan
ID: 1, Name: Saniya Bagwan, Email: san@gmail.com, Role: SDE, Salary: 130000.0
```

Updating

```
===== SMART HR SYSTEM =====
1. Add Employee
2. Display All Employees
3. Search Employee
4. Update Employee Role
5. Delete Employee
6. Display Unique Roles
7. Exit
Enter choice: 4
Enter ID: 3
Enter new role: Data analyst
Updated successfully!
```

Displaying Unique Roles

```
===== SMART HR SYSTEM =====
1. Add Employee
2. Display All Employees
3. Search Employee
4. Update Employee Role
5. Delete Employee
6. Display Unique Roles
7. Exit
Enter choice: 6
Unique Roles: [Data analyst, SDE, Contracter]
```

Deleting

```
===== SMART HR SYSTEM =====
1. Add Employee
2. Display All Employees
3. Search Employee
4. Update Employee Role
5. Delete Employee
6. Display Unique Roles
7. Exit
Enter choice: 5
Enter ID: 2
Deleted successfully!

===== SMART HR SYSTEM =====
1. Add Employee
2. Display All Employees
3. Search Employee
4. Update Employee Role
5. Delete Employee
6. Display Unique Roles
7. Exit
Enter choice: 2
ID: 1, Name: Saniya Bagwan, Email: san@gmail.com, Role: SDE, Salary: 130000.0
ID: 3, Name: Manasvi Gore, Email: man@gmail.com, Role: Data analyst, Salary: 31200.0
```